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DEPARTMENT OF PHYSICS

PHYSICS LAB 7

Fall 2023-2024

LAB REPORT ON

Determination of the time constant of an RC circuit.

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Submission: 07/12/2023

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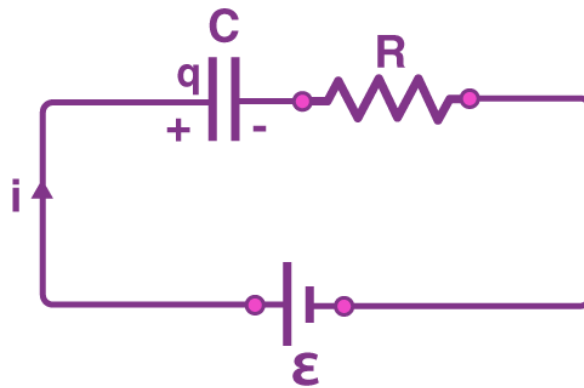
1. Theory

A resistor-capacitor circuit (RC Circuit) is an electrical circuit consisting of passive components like resistors and capacitors, driven by the current source or the voltage source.

The capacitor stores energy, and the resistor connected to the circuit controls the rate of charging or discharging.

The charging and discharging of the capacitor is not an instant process but takes some time. If the resistor and capacitor are connected in series, the capacitor charges gradually through the resistor until the voltage across the resistor is equal to the supply voltage.

Charging of Capacitor



Let us consider that a fully discharged capacitor is connected to the circuit, and the switch is open. The initial condition of the circuit is time (t) = 0, charge (q) = 0 and current (i) = 0. When the switch is closed, the time will start from $t = 0$, and the current starts flowing through the capacitor and the resistor. An initially uncharged capacitor can be assumed to be a connecting wire just after the circuit is completed.

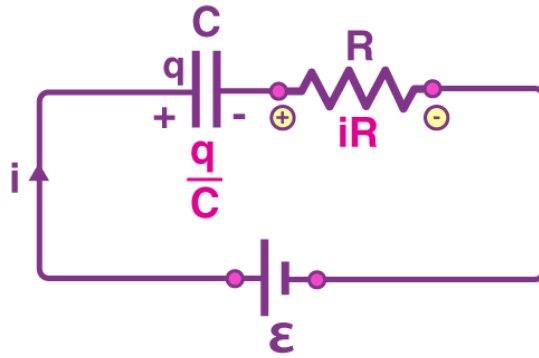
At time $t = 0$, the potential difference across the capacitor is zero and continues to be equal to zero just after the time $t = 0$.

The current flowing at this time is called the charging current, and it is calculated using Ohm's law. The value of the current is,

$$i = i_0 = \varepsilon/R.$$

Here, ε is the *emf* of the cell.

R is the resistance of the resistor.



As time progresses, the capacitor charges up, and a potential difference develops across the capacitor. Let us consider at any time “t”, the charge across the capacitor is “q”, and the current in the circuit is “i”. The potential difference across the capacitor is $\frac{q}{C}$, and the potential difference across the resistor is iR .

We find from Kirchhoff’s loop theorem:

$$\varepsilon - iR - \frac{q}{C} = 0$$

$$\varepsilon = iR + \frac{q}{C} \quad [i \text{ \& } q = \text{two variables, } i = \frac{dq}{dt}]$$

$$\varepsilon = R \frac{dq}{dt} + \frac{q}{C} \dots \dots \dots (1)$$

$$\text{And, } q = q_0 \left(1 - e^{\frac{-t}{RC}}\right)$$

$$CV_c = C\varepsilon(1 - e^{\frac{-t}{RC}}) \quad CV_c = C\varepsilon(1 - e^{\frac{-t}{RC}}) \quad) \quad \text{since, } [q(t) = CV_c(t)] \text{ \& } [q_0 = C\varepsilon]$$

$$\ln(1 - \frac{V_c}{\varepsilon}) = (-\frac{1}{RC})t \dots \dots \dots (2)$$

Discharging of capacitor:

Assume now that the capacitor is fully charged to a potential V_0 equal to the emf ε of the battery. At a new time, $t = 0$, switch is thrown from A to B so that the capacitor can discharge through resistance R . Now the charge $q(t)$ on the capacitor and the

current $i(t)$ through the discharge loop of capacitor and resistance now vary with time.

With no battery in the discharge loop, $\mathcal{E} = 0$, Eq. (1) becomes using Kirchhoff's loop theorem:

$$iR + \frac{q}{C} = 0$$

$$R \frac{dq}{dt} + \frac{q}{C} = 0 \quad [i = \frac{dq}{dt}]$$

$$\text{And, } q = C \int_0^t \mathcal{E} e^{-\frac{t}{RC}} dt$$

$$CV_c = C \mathcal{E} e^{-\frac{t}{RC}} \quad \text{since, } [q(t) = CV_c(t)] \text{ \& } [q_0 = C\mathcal{E}]$$

$$\ln\left(\frac{V_c}{\mathcal{E}}\right) = \left(-\frac{1}{RC}\right)t \dots \dots \dots (3)$$

Comparing Eq. 2 and 3 with $y = mx+c$ and plotting a graph of " $\ln(1 - \frac{V_c}{\mathcal{E}})$ vs t " for charging and " $\ln(\frac{V_c}{\mathcal{E}})$ vs t " for discharging, we get the value of τ as $\tau = -\frac{1}{m}$, where m is the slope of the graphs.

Time Constant

The time constant (τ) = RC

2. Apparatus

- Power supply,
- Circuit board,
- Resistor,
- Capacitor,
- Multi meter,
- Stopwatch and
- Connecting wires.

3. Procedure

- An RC circuit was constructed on the circuit board as per the circuit diagram provided.
- Upon applying a sufficient voltage from the power supply, the charging of the capacitor was observed, and the voltage differences across the capacitor were noted with time.
- Following the charging phase, the power supply was disconnected from the circuit, allowing the observation of the capacitor discharging over time.

Simultaneously, the voltage differences across the capacitor were noted throughout the discharging process.

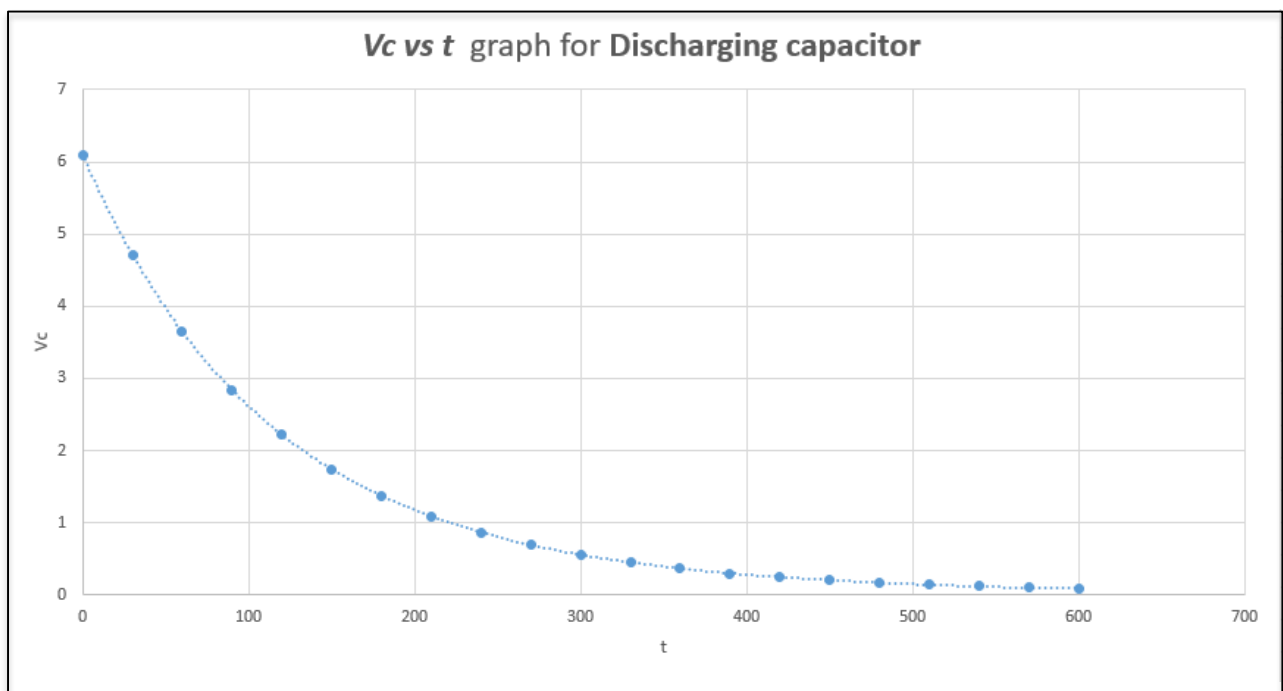
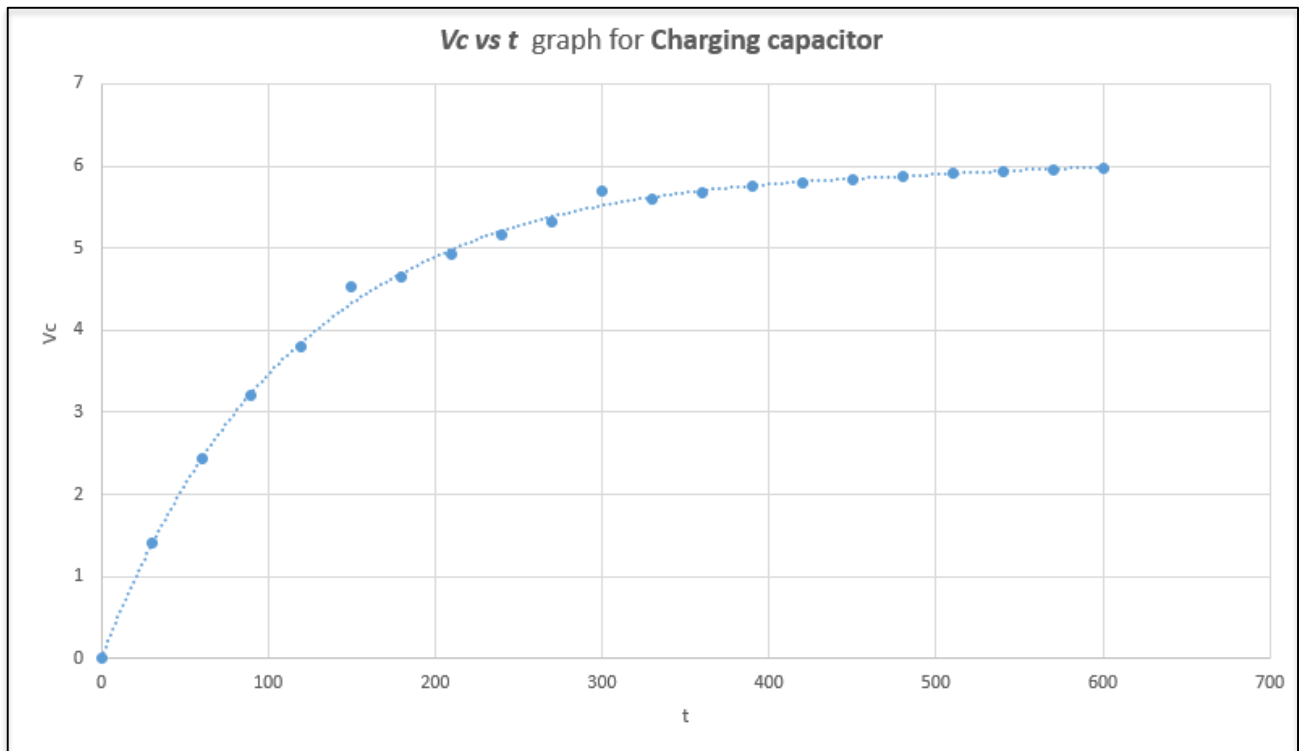
4. Experimental Data:

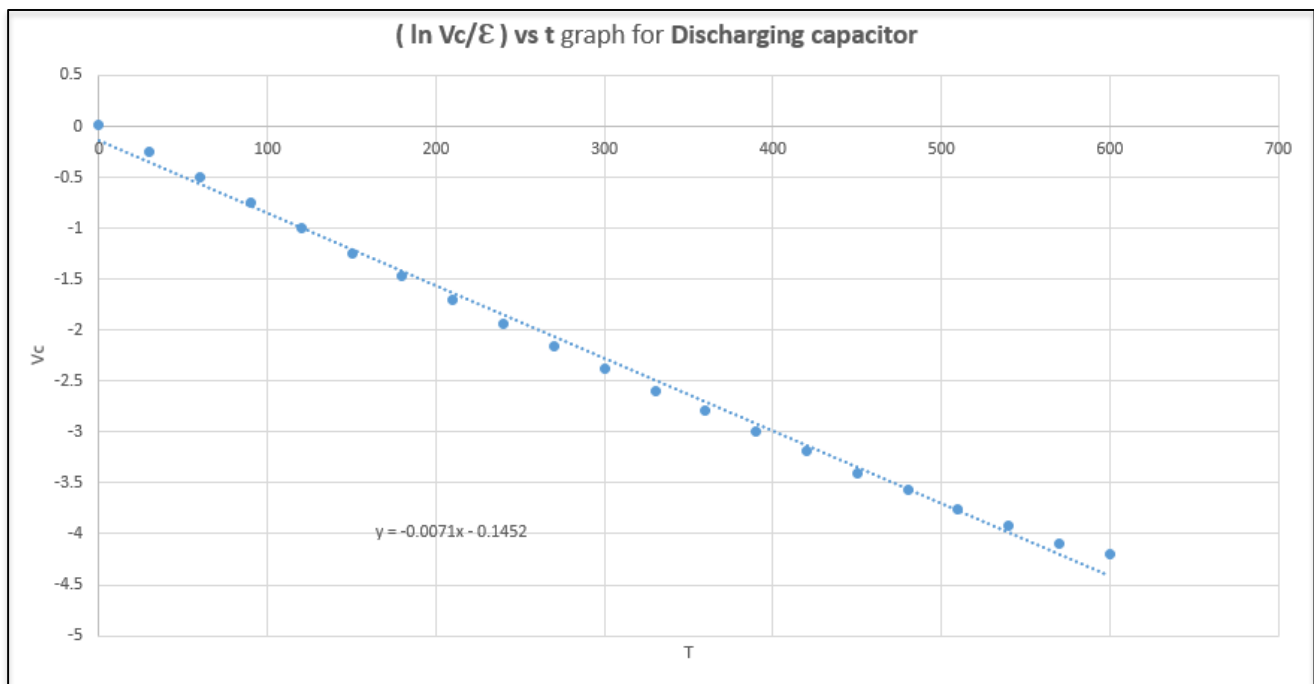
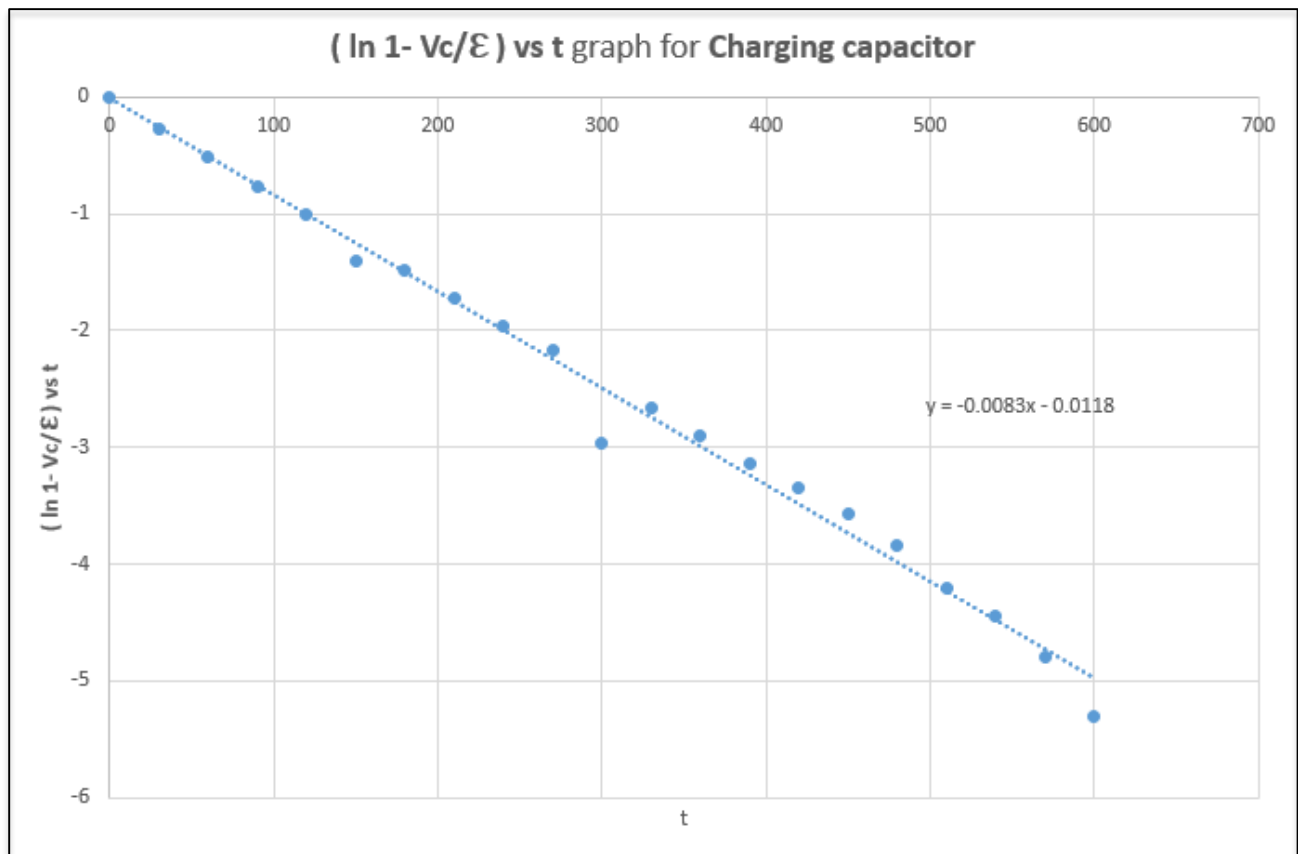
Table 1: Charging & Discharging of an RC circuit.

Maximum potential difference ($\mathcal{E}$) = 6.01 volts

Time (s)	Charging capacitor		Discharging capacitor	
	V_c (volt)	$\ln \left(1 - \frac{V_c}{\mathcal{E}}\right)$	V_c (volt)	$\ln \frac{V_c}{\mathcal{E}}$
0	0	0	6.1	0.015
30	1.4	-0.265	4.7	-0.246
60	2.43	-0.518	3.66	-0.496
90	3.2	-0.760	2.84	-0.750
120	3.8	-1.001	2.23	-0.991
150	4.52	-1.395	1.73	-1.245
180	4.64	-1.479	1.38	-1.471
210	4.93	-1.716	1.09	-1.707
240	5.16	-1.956	0.87	-1.933
270	5.32	-2.164	0.69	-2.164
300	5.7	-2.965	0.56	-2.373
330	5.59	-2.661	0.45	-2.592
360	5.68	-2.902	0.37	-2.788
390	5.75	-3.140	0.3	-2.997
420	5.8	-3.354	0.25	-3.180
450	5.84	-3.565	0.2	-3.403
480	5.88	-3.834	0.17	-3.565
510	5.92	-4.201	0.14	-3.759
540	5.94	-4.453	0.12	-3.914
570	5.96	-4.789	0.1	-4.096
600	5.98	-5.300	0.09	-4.201

5. Graphs





6. Calculation:

Charging a capacitor

Theoretically:

$$\begin{aligned}\text{Time constant, } \tau &= RC \\ &= 47 \times 10^3 \times 2200 \times 10^{-6} \\ \tau &= 103.4\end{aligned}$$

Experimentally:

From the graph: Slope, $m = -\frac{1}{\tau}$

$$\tau = -\frac{1}{m} = \frac{-1}{-0.0083} = 120.48$$

$$\text{Error of } \tau = \frac{\tau_{th} - \tau_{ex}}{\tau_{th}} \times 100\% = \frac{120.48 - 103.4}{103.4} \times 100\% = 16.52\%$$

Discharging a capacitor

Theoretically:

$$\begin{aligned}\text{Time constant, } \tau &= RC \\ &= 47 \times 10^3 \times 2200 \times 10^{-6} \\ \tau &= 103.4\end{aligned}$$

Experimentally:

From the graph: Slope, $m = -\frac{1}{\tau}$

$$\tau = -\frac{1}{m} = \frac{-1}{-0.0071} = 140.85$$

$$\text{Error of } \tau = \frac{\tau_{th} - \tau_{ex}}{\tau_{th}} \times 100\% = \frac{140.85 - 103.4}{103.4} \times 100\% = 36.21\%$$

7. Result:

Table 7.2: Values of time constant, τ .

From the graphs		Theoretically: $\tau (=RC)$
Process	Experimentally: $\tau = -\frac{1}{m}$	
Charging	120.48	103.4
Discharging	140.85	103.4

8. Discussion :

Capacitor and Capacitance: A capacitor is an electronic component designed to store and release electrical energy. It consists of two conductive plates separated by a non-conductive material known as a dielectric. Capacitance (C) is a measure of a capacitor's ability to store charge per unit voltage. It's determined by the area of the plates, the distance between them, and the properties of the dielectric.

Charging and Discharging: When a capacitor is connected in a circuit:

Charging: If connected to a voltage source through a resistor, current flows into the capacitor, gradually storing charge until it reaches the voltage of the source. Initially, the charging rate is high, but it diminishes over time.

Discharging: If a charged capacitor is disconnected from the voltage source and connected across a resistor, it discharges its stored energy as current flows through the resistor. Initially, the discharging rate is high, but it decreases over time until the capacitor is fully discharged.

Significance of Time Constant in an RC Circuit:

The time constant (τ) in an RC circuit is the product of the resistance (R) and capacitance (C) in the circuit, $\tau = R \times C$.

It represents the time required for the voltage across the capacitor to either charge up to approximately 63% of the maximum voltage during charging or discharge down to approximately 37% of its initial voltage during discharging.

The time constant is crucial as it characterizes the rate at which the capacitor charges or discharges in response to changes in the circuit conditions.

Obtaining Time Constant of an RC Circuit:

The time constant (τ) of an RC circuit can be calculated by multiplying the resistance (R) and capacitance (C) values used in the circuit, $\tau = R \times C$.

Voltage across the Capacitor in an RC Circuit vs. Time:

Charging: Initially, the voltage across the capacitor is zero. As it charges, the voltage gradually increases, approaching the source voltage. The curve follows an exponential rise towards the maximum voltage, following the equation $V_c = V_s \times (1 - e^{-\frac{t}{RC}})$, where V_c is the voltage across the capacitor, V_s is the source voltage, t is time, and RC is the time constant.

Discharging: Initially, the voltage across the capacitor is its fully charged voltage. As it discharges, the voltage gradually decreases towards zero exponentially, following the equation $V_c = V_0 \times e^{-\frac{t}{RC}}$, where V_0 is the initial voltage across the capacitor.

9. Resources

For further understanding students may go through the following resources:

- Fundamental of Physics (10th Edition): Capacitor (Chapter 25, page 717-721), RC circuit (Chapter 27, page 788-791).
- <https://byjus.com/jee/rc-circuit/>
- Video Links:
 - ✓ https://www.youtube.com/watch?v=f_MZNsEqyQw
 - ✓ (4) 22 - Circuits - Time constant of an RC circuit - YouTube